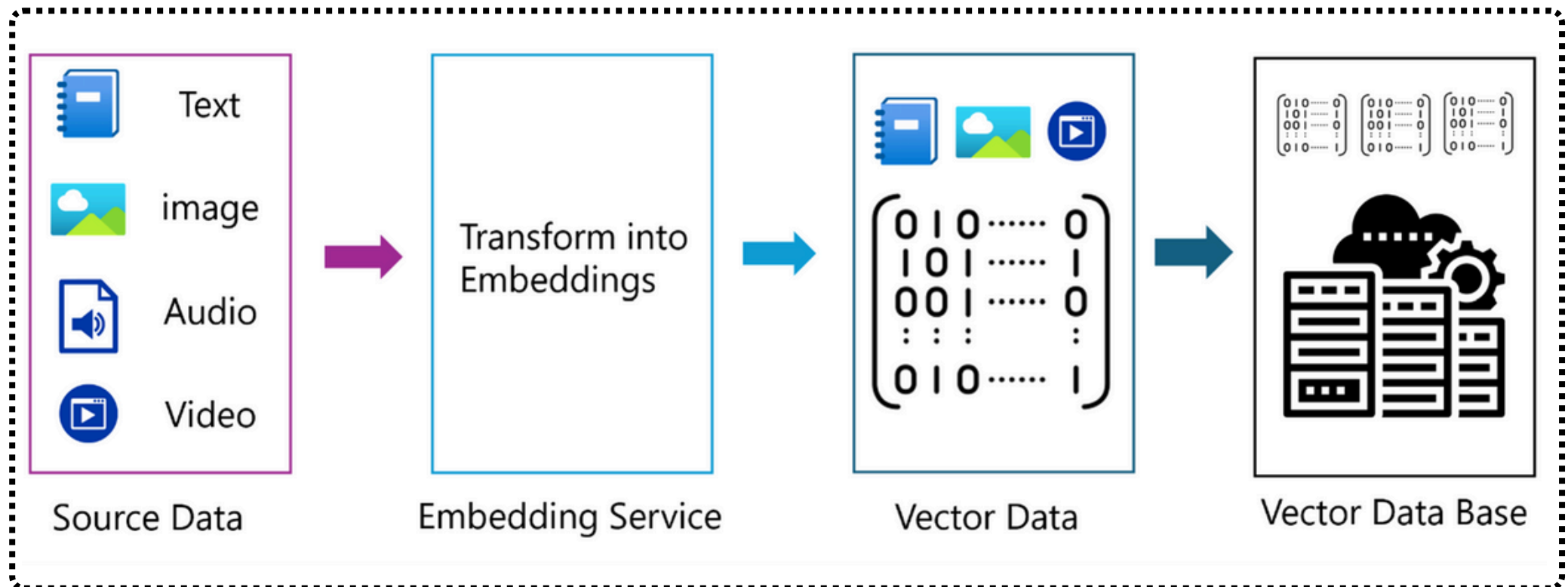


# A Complete Guide to Vector Databases

## Complete Notes



### Dedicated vector databases

### Databases that support vector search

| Open source<br>(Apache 2.0 or MIT license) | Dedicated vector databases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Databases that support vector search                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            |  <b>chroma</b><br> <b>marqo</b><br> <b>vespa</b><br> <b>qdrant</b><br> <b>LanceDB</b><br> <b>Milvus</b> |  <b>OpenSearch</b><br> <b>ClickHouse</b><br> <b>PostgreSQL</b><br> <b>cassandra</b> |
| Source available<br>or commercial          |  <b>Weaviate</b><br> <b>Pinecone</b>                                                                                                                                                                                                                                                                                                                                                                                                                |  <b>elasticsearch</b><br> <b>redis</b><br> <b>[ROCKSET]</b><br> <b>SingleStore</b>  |

# 1. Why Vector Databases Exist

Traditional databases are good at answering questions like:

- Find users with age = 30
- Find orders where price > 100
- Find products where category = shoes

But they completely fail at questions like:

- Find text similar to "user is angry about late delivery"
- Find images similar to this photo
- Find documents related to "LLM hallucination"
- Find code similar to this function

These are **semantic** queries.

They are about **meaning**, not exact values.

That is why **vector databases exist**.

They store data as **vectors** so that **similar things are close together** in mathematical space.

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## 2. What is a Vector?

A **vector** is just a list of numbers.

Example:

```
csharp
```

[Copy code](#)

```
[0.21, -0.98, 1.34, 0.07, ...]
```

In ML, vectors represent meaning.

Text, images, audio, code all get converted into vectors using **embedding models**.

| Data             | Embedding                |
|------------------|--------------------------|
| "I love dogs"    | [0.12, -0.44, 1.2, ...]  |
| "I like puppies" | [0.11, -0.46, 1.18, ...] |
| "I hate traffic" | [-0.91, 0.02, -1.4, ...] |

Notice how similar meanings produce similar vectors.

### 3. What is a Vector Database?

A **vector database** stores:

- Vectors
- Metadata
- Original content

and supports:

- Fast similarity search
- Filtering
- Scalable retrieval

Instead of SQL like:

```
java
```

[Copy code](#)

```
WHERE age = 30
```

You query using:

```
cpp
```

[Copy code](#)

```
Find top 10 vectors closest to this vector
```

### 4. What Makes Vector Search Work

Vectors live in a high dimensional space.

A text embedding might have 768 or 1536 dimensions.

Similarity is computed using distance functions:

| Metric             | Meaning                |
|--------------------|------------------------|
| Cosine similarity  | Angle between vectors  |
| Euclidean distance | Straight line distance |
| Dot product        | Directional similarity |

Cosine similarity is most common for text.

## 5. Why You Cannot Use SQL for This

SQL databases:

- Are row based
- Use indexes on fixed columns
- Work with exact values

Vector search requires:

- High dimensional nearest neighbor search
- Approximate search
- In memory optimized indexes

Trying this in PostgreSQL or MySQL will be slow and painful.

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## 6. Core Architecture of a Vector Database

A vector DB has 4 main layers.

### 1. Ingestion Layer

Takes:

- Raw data
- Vectors
- Metadata

### 2. Indexing Layer

Builds ANN indexes like:

- HNSW
- IVF
- PQ

### 4. Query Engine

Accepts:

- A vector
- Filters
- Top K

Returns:

- Most similar items

This allows fast similarity search.

### 3. Storage Layer

Stores:

- Vectors
- Metadata
- IDs

## 7. ANN Search Explained Simply

ANN means **A**pproximate **N**earest **N**eighbor.

Exact search checks every vector.

That is impossible at scale.

ANN uses graph and clustering tricks to search only the promising part of space.

HNSW builds a layered graph where similar vectors connect.

IVF clusters vectors and only searches relevant clusters.

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## 8. How Data Flows

Let's say you want to build an AI document search.

### Step 1: Chunk your documents

Split PDFs, docs, pages into chunks.

### Step 2: Convert text to vectors

Using an embedding model.

```
arduino
```

[Copy code](#)

```
"This product is bad" → [0.34, -1.2, ...]
```

### Step 3: Store in vector DB

You store:

```
bash
```

[Copy code](#)

```
id: 1
vector: [0.34, -1.2, ...]
text: "This product is bad"
metadata: {source: "review", date: "2024-01-01"}
```



Step 4: User asks a question

"Why are customers unhappy?"

Convert the question into a vector.

Step 5: Similarity search

Find nearest vectors.

Step 6: Send results to LLM

Now the LLM answers using real data.

This is called **RAG**.

9. Why Vector DBs Power LLMs

LLMs do not remember your data.

Vector databases give LLMs memory.

This enables:

- Chat with PDFs
- Customer support bots
- Enterprise search
- Code assistants
- Recommendation systems

Without vector DBs, LLMs hallucinate.

13. Hybrid Search

Best systems do:

```
nginx

vector_score * 0.7 + keyword_score * 0.3
```

This gives accuracy and semantic power.

10. Popular Vector Databases

| Name     | Notes                  |
|----------|------------------------|
| Pinecone | Fully managed          |
| Weaviate | Open source + cloud    |
| Milvus   | High scale open source |
| Qdrant   | Fast, Rust based       |
| FAISS    | Library, not DB        |
| Chroma   | Local dev              |
| pgvector | Postgres plugin        |

11. What Makes a Good Vector Database

You want:

- Fast ANN search
- Filtering on metadata
- Horizontal scaling
- Persistence
- Hybrid search (text + vectors)
- High recall

12. Vector DB vs Search Engine

| Search Engine      | Vector Database  |
|--------------------|------------------|
| Keyword based      | Meaning based    |
| BM25               | Embeddings       |
| Exact matches      | Semantic matches |
| Misses paraphrases | Handles synonyms |

Modern systems combine both.

## 14. Cost Model

Vector DBs are expensive because:

- Vectors are large
- RAM heavy
- Indexing is complex

Cost drivers:

- Number of vectors
  - Dimension size
  - Replication
  - Query volume
- 

## 15. When You Should Use One

Use a vector DB if you need:

- Semantic search
- Recommendation engines
- AI agents with memory
- Document QA
- Similarity matching
- Fraud detection
- Image or audio search

Do not use one for simple CRUD.

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## 16. The Mental Model

Think of a vector DB as:

A Google Maps of meaning.

Queries are locations.

Data points are places.

Nearby means similar.